One Percent of Which Comparisons? Exposure and Uncertainty in Pairwise Leaderboard Stability

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Abstract

Small fractions of edited comparisons can change a fitted leaderboard, but the fraction alone hides who is affected and whether the original leader was distinguishable. We audit this interpretation using fixed public snapshots of crowdsourced model preferences, curated human and GPT-4 judgments, and three NBA seasons. Outcome reversals are selected by standard influence scores and validated by full Bradley–Terry refitting; random edits and influence-based deletion are controls. We report both the global edit fraction and the maximum fraction of any entrant's comparisons touched. At a nominal 1% global budget, uncapped outcome reversals switch the original top-pair ordering in 4 of 6 full cohorts. The same edits can touch substantially more than 1% of one entrant's record. Requiring each entrant's exposure to stay within 1% leaves only 1 observed reversal, in an Arena cohort whose original margin is already compatible with zero. Integer feasibility accounts for much of the NBA contrast; it is not evidence of certified robustness. Matched human–judge panels and specification checks further separate initial score separation from perturbation response. The contribution is an exposure-aware, uncertainty-explicit empirical audit, not a new influence method, optimal attack, or causal comparison of evaluation designs.

1 Introduction

Pairwise leaderboards compress heterogeneous evidence into a simple ordering. A model can be advertised as first, even when its fitted advantage is smaller than ordinary sampling variation, and a change to a small fraction of the entire comparison archive can be described as a small disturbance to that model's evidence. Neither interpretation follows automatically. The total archive, an entrant's incident comparisons, and the experimental unit need not be interchangeable denominators.

Existing research already establishes that influential preference deletions can change Bradley–Terry (BT) model rankings, including on Arena and MT-Bench [3]. Influence-function tools [4] and approximate maximum influence perturbations (AMIP) [1] provide efficient ways to find such subsets. Therefore, showing another low-budget flip or applying an inverse Hessian to a BT margin is not our contribution. Our question is narrower: *what does a global fragility number conceal about individual exposure, feasible perturbations, and original uncertainty?*

This matters even without a malicious adversary. Random errors, removal of questionable votes, systematic changes to a judging policy, and targeted relabeling are different operations. Concentrating changes on a single entrant can use a small global fraction while modifying a much larger share of that entrant's evidence. Conversely, an integer interpretation of a small per-entrant percentage

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may admit almost no edits in a short schedule. Finally, a point-order reversal need not overturn an originally supported separation.

We analyze three observational evaluation designs: crowdsourced Arena preferences, curated MT-Bench judgments (human and model judge), and fixed-season professional basketball outcomes. These designs differ in participant count, comparison density, ties, task sampling, and dependence. They are not randomized treatments. Our results establish conditional differences between the selected snapshots, not an effect of replacing humans with model judges or sports schedules with crowdsourcing.

The audit has three components. First, each exact-refit witness is accompanied by a global fraction, realized per-entrant exposure, and actual edit count. Second, strict incident-exposure caps are evaluated alongside unconstrained influence selection, with integer-linear optimization checking the capped surrogate and feasibility bounds separating arithmetic restrictions from failed searches. Third, original cluster-bootstrap uncertainty and random-edit variation are kept distinct from adversarial response; a common-cell MT comparison controls the comparison schedule without equating aggregate cells with individual votes.

The resulting message is conditional rather than uniformly alarmist. Global low-budget flips occur, but near-ties and concentrated exposure change their meaning. The strict-cap comparison is informative only together with the feasible number of edits. No failed search in this paper is a robustness certificate. All modifications are offline counterfactuals to published records, not interventions on a live leaderboard.

2 Related work

Leaderboards and judging designs. Arena collects user preferences over anonymous model responses and estimates model scores and uncertainty [2]. MT-Bench supplies a curated question set and human and GPT-4 assessments, while documenting judge biases and consistency issues [6]. We reuse their published labels; no new model responses or paid judge calls are generated. Our paired analysis uses common question–turn–model-pair cells rather than assuming that the full human and GPT-4 releases have the same sampling pattern.

Influence and finite-sample robustness. Koh and Liang [4] use inverse-Hessian derivatives to trace the effect of changing observation weights. Broderick et al. [1] distinguish exact maximum perturbations from first-order approximations, and stress that refitting a candidate deletion can demonstrate non-robustness but cannot certify robustness when the candidate fails. They also distinguish sign changes, significance changes, and statistically significant opposite conclusions. We retain these distinctions: our standardized effects are reference-scale quantities, not new significance tests.

Ranking sensitivity and manipulation. Huang et al. [3] already apply AMIP to BT score differences and study deletion sensitivity on Arena and MT-Bench, including human versus model judgments. We credit that core application rather than presenting it as new. Min et al. [5] study practical vote rigging in Arena, including identification and attack feasibility. Our fixed-record outcome reversals assume knowledge and editing access that a real user may not have; we do not reproduce a live manipulation system. The narrower addition is a cross-domain audit of the edit denominator and strict endpoint constraints, paired with uncertainty and edit-unit provenance. This is not an exhaustive novelty claim: the task imposed a bounded literature-access protocol, and only explicitly permitted sources were consulted (Appendix H.1).

3 Estimands and audit procedures

3.1 Fitted score margin

A record r compares distinct entrants a_r, b_r, with response $y_r \in [0, 1]$: one means a_r wins, zero means b_r wins, and one half denotes a tie. Orientations are canonicalized to $a_r < b_r$, replacing y_r by

$1 - y_r$ when necessary. Let $x_r = e_{a_r} - e_{b_r}$. We fit

$$\widehat{\theta} = \arg\min_{\theta \in \mathbb{R}^P} \sum_{r=1}^{N} \left\{ \log(1 + \exp(x_r^\top \theta)) - y_r x_r^\top \theta \right\} + \frac{\lambda}{2} \|\theta\|_2^2. \tag{1}$$

The loss is a *sum*, not a mean. A positive ridge, fixed at $\lambda = 0.1$ before outcome inspection, makes every fitted solution finite and unique; its coordinates sum to zero. Ties are fractional outcomes, not a separately estimated tie model. Sensitivities to ridge and tie handling are reported rather than tuned away.

The original full-cohort leader ℓ and runner-up j define a fixed contrast $c = e_\ell - e_j$ and margin $m_0 = c^\top \widehat{\theta} > 0$. The same pair is tracked after perturbation, under bootstrap resampling, and under altered model specifications. Other challengers may incidentally become first, but we do not optimize against every challenger. This conditional target avoids silently changing the question after each edit; it does not remove the selection bias of choosing the original top pair.

3.2 Edit units and exposure

For a set S of distinct records, the primary operation reverses outcomes: $y'_r = 1 - y_r$ for $r \in S$. Exact ties are unchanged and ineligible. Counts, competitors, and the observed comparison schedule remain fixed. A deletion control instead removes those records from the likelihood, without changing λ. These are different perturbation models, and no common numerical budget makes them identical.

Let n_i be entrant i's original incident-record count and $k_i(S)$ its edited incident count. We report

$$\alpha(S) = \frac{|S|}{N}, \qquad \rho(S) = \max_{i: n_i > 0} \frac{k_i(S)}{n_i}, \qquad \alpha(S) = \sum_i \frac{n_i}{2N} \frac{k_i(S)}{n_i} \le \rho(S). \tag{2}$$

Thus the global fraction is an exposure-weighted average, not the maximum entrant disturbance. On a balanced schedule $n_i \simeq 2N/P$, concentrating k edits on one entrant can give $k/n_i \simeq \alpha P/2$. Participant count alone can therefore change the interpretation of the same global fraction.

For nominal budget α, the unconstrained procedure uses at most $B = \lfloor \alpha N \rfloor$ edits. The *strict-cap* procedure additionally requires $k_i(S) \le \lfloor \alpha n_i \rfloor$ for every entrant. This is a deliberately stringent audit restriction, not a model of all plausible noise or adversaries. Since every edit consumes two endpoint slots,

$$|S| \le \left\lfloor \frac{1}{2} \sum_i \lfloor \alpha n_i \rfloor \right\rfloor. \tag{3}$$

This structural count bound is valid even before considering which eligible edges exist. Reporting only the nominal budget can obscure such infeasibility.

3.3 Influence selection, controls, and exact refitting

Let H be the Hessian of Eq. (1) at the original fit and $p_r = \mathrm{sigmoid}(x_r^\top \widehat{\theta})$. The first-order predicted margin reductions for reversing and deleting record r are respectively

$$s_r^{\mathrm{flip}} = c^\top H^{-1} x_r (2y_r - 1), \qquad s_r^{\mathrm{del}} = c^\top H^{-1} x_r (y_r - p_r). \tag{4}$$

Appendix A gives the signed derivative calculation. Sorting beneficial scores maximizes the uncapped first-order objective. Under caps, a deterministic greedy scan accepts a record only when both endpoint capacities remain. Scores are not recomputed in the primary procedure; harmful or zero-score edits do not fill spare capacity. Stable row order resolves ties.

We also optimize the *integer linear surrogate* at 1% and 5%, aggregating identical endpoint/outcome classes into bounded integer variables. This verifies the surrogate objective under caps, not the non-linear BT objective. An uncapped batch-adaptive procedure, with at most ten score recomputations, tests sensitivity to the static approximation. Every selected set is then evaluated by minimizing the full perturbed objective, not by trusting its linear prediction.

Uniform subsets of non-tie records provide the main random-reversal control. Additional controls use the same count as the capped greedy set, or a random-priority greedy scan satisfying the caps. The

Table 1: Full cohorts and original fixed top-pair margins. Intervals are 95% cluster-bootstrap percentile intervals with 300 replicates. P: entrants. Ties include GPT-4's inconsistent judgments under the main convention. MT full releases are not a matched panel.

Cohort	N	P	Clusters	Tie%	m_0	Bootstrap interval
Arena	57,477	64	51734	30.9	0.004	[-0.103, 0.106]
MT human	3,355	6	80	23.2	0.378	[0.108, 0.661]
MT GPT-4	2,400	6	80	25.0	0.820	[0.544, 1.129]
NBA 2013	1,314	30	1314	0.0	0.228	[-0.409, 0.845]
NBA 2014	1,319	30	1319	0.0	0.295	[-0.340, 0.930]
NBA 2015	1,311	30	1311	0.0	0.744	[0.159, 1.386]

latter is *not* uniform over feasible sets, and it can underfill. Influence-based deletion is an AMIP-style control credited to prior work, not a newly introduced baseline. A refitted negative margin is a feasible reversal witness and an upper bound on the minimum necessary edit budget. Failure to obtain one says nothing about the best possible unsearched set.

3.4 Effect size and uncertainty

The primary continuous endpoint at nominal 1% is the exact margin reduction $m_0 - m_S$. We also report $D = (m_0 - m_S)/s_0$, where s_0 is the SD of the original fixed-pair cluster-bootstrap margins. This standardizes a change against ordinary original-data variation; D is *not* a post-edit z-test. Pair reversal $m_S < 0$ and incidental winner change are separate outcomes.

Three different variations are kept apart. First, 300 original-data cluster-bootstrap replicates give s_0 (sample SD, divisor 299) and percentile intervals. Second, 100 random-edit seeds per budget give empirical 2.5–97.5% subset ranges, and Wilson intervals for Monte Carlo reversal probabilities. These ranges do not estimate population sampling uncertainty. Third, a post-outcome exploratory extension repeats fitting, set selection, and refitting within 300 cluster-bootstrap samples at 1%; its percentile ranges describe the *resampled procedure*, not the effect of a fixed set. All percentages use integer floors. No hypothesis-testing sweep is performed, and none of the reported intervals is presented as simultaneous or selection-adjusted coverage.

4 Data and experimental setup

Fixed public snapshots. We use the supplied Arena55k release [2], both MT-Bench judgment files [6], and FiveThirtyEight's NBA historical records. Table 1 reports the six full cohorts. All retained comparison graphs are connected, with unique record IDs. NBA seasons were fixed to 2013–2015 before inspecting scores, include regular season and playoffs, and use franchise IDs and one row per game ($_iscopy = 0$, NBA league). Historical Elo columns are never used. This is a descriptive static BT analysis, not a reconstruction of an official time-varying basketball ranking.

Arena clustering uses the exact prompt-string hash, linking repeated prompt strings but not unknown users or paraphrases. MT resampling clusters all turns, pairs, and judges for a question. NBA resampling uses games; it does not capture serial or shared-team dependence. These are explicit uncertainty approximations, not universal coverage guarantees.

MT human records include both `expert_` and `author_` judge labels. The main human cohort includes both, with source-restricted checks in the appendix. GPT-4's `tie (inconsistent)` category is mapped to 0.5 in the main analysis, as are explicit ties; an exclusion check separates this convention from genuine indifference. There are no duplicate canonical question–turn–pair–judge keys in either release.

Matched comparison schedule. For each MT question, turn, and unordered model pair, we average available human responses, then intersect cells with GPT-4 and assign each cell unit weight. This gives 1814 common cells per evaluator. Some human cell responses are fractional but not exactly one half. Reversing a cell reverses its aggregate response, potentially representing several individual judgments; it is not an equally costly vote edit. The full and matched cohorts overlap and are

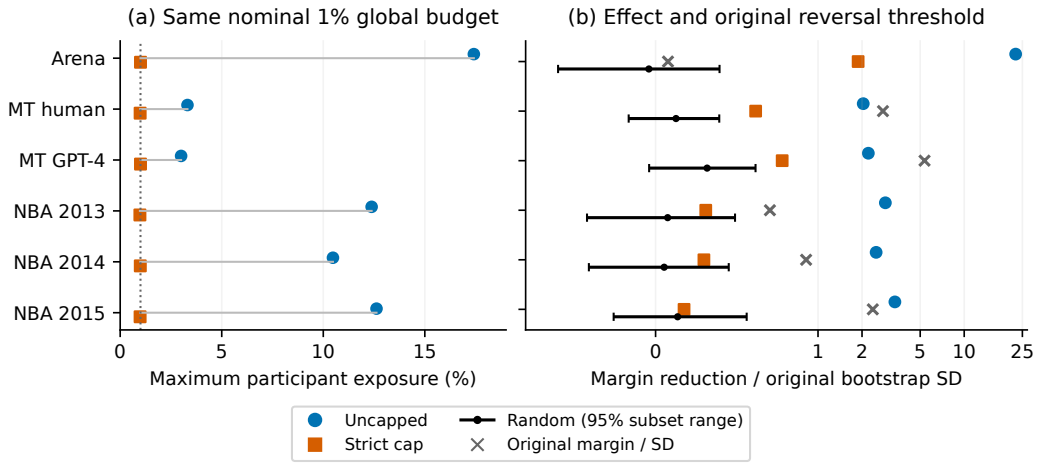


Figure 1: Primary 1% audit on six full cohorts. (a) Realized maximum entrant exposure, not the nominal global fraction. (b) Exact margin reductions in original-bootstrap SD units (symlog axis, linear within one unit); crosses mark each original margin/SD, beyond which the fixed pair reverses. Black whiskers are random-subset percentile ranges, not confidence intervals for deterministic attacks. Capped counts can be much smaller (Table 2).

Table 2: Nominal 1% outcome-reversal budget. U: uncapped; C: strict capped greedy. k: actual edits; ρ_U: maximum entrant exposure; D: exact reduction/original bootstrap SD. Last column is reversal frequency over 100 random subsets, not a worst-case rate; Wilson intervals are in Table 5.

Cohort	k_U	$100\rho_U$	D_U	k_C	D_C	Random flip
Arena	574	17.4	22.49	555	1.88	35%
MT human	33	3.3	2.04	29	0.62	0%
MT GPT-4	24	3.0	2.21	24	0.78	0%
NBA 2013	13	12.4	2.89	1	0.31	1%
NBA 2014	13	10.5	2.50	2	0.30	0%
NBA 2015	13	12.6	3.36	1	0.18	0%

not independent replications. Question-resampling draws are synchronized across panels, and the common original top pair is GPT-4 versus Claude-v1.

Protocol and resources. The pre-outcome plan fixes $\lambda = 0.1$, the 1% endpoint, budgets $0, 0.1, 0.5, 1, 2, 5, 10\%$, bootstrap count 300, random count 100, and base seed 20261104. Ridge 0.01 and 1, tie exclusion, and adaptive selection are sensitivity checks. Supplementary controls added after the numerical pilot and the post-outcome bootstrap are marked separately in the execution log. Newton optimization uses pair sufficient statistics; finite-difference and analytical tests validate the solver and both derivatives. CPU execution is single-threaded for numerical libraries; no GPU, paid API, or new annotation is used. Versions, runtimes, data hashes, complete commands, and the failure log are included with the code. Appendix G specifies the reproduction path.

5 Results

5.1 A global fraction can conceal concentrated exposure

At the primary budget, static uncapped reversal changes the original top-pair ordering in Arena and each of the three NBA seasons, but not in either full MT cohort (Table 2). In Arena, 574 reversals touch up to 17.41% of one model's incident record. Strict caps permit 555 selected edits and reduce the standardized effect from 22.494 to 1.881. The two procedures use similar global counts but different exposure allocations. Capping still reverses Arena's tiny original margin; the contrast is an effect-size difference, not a universal defense.

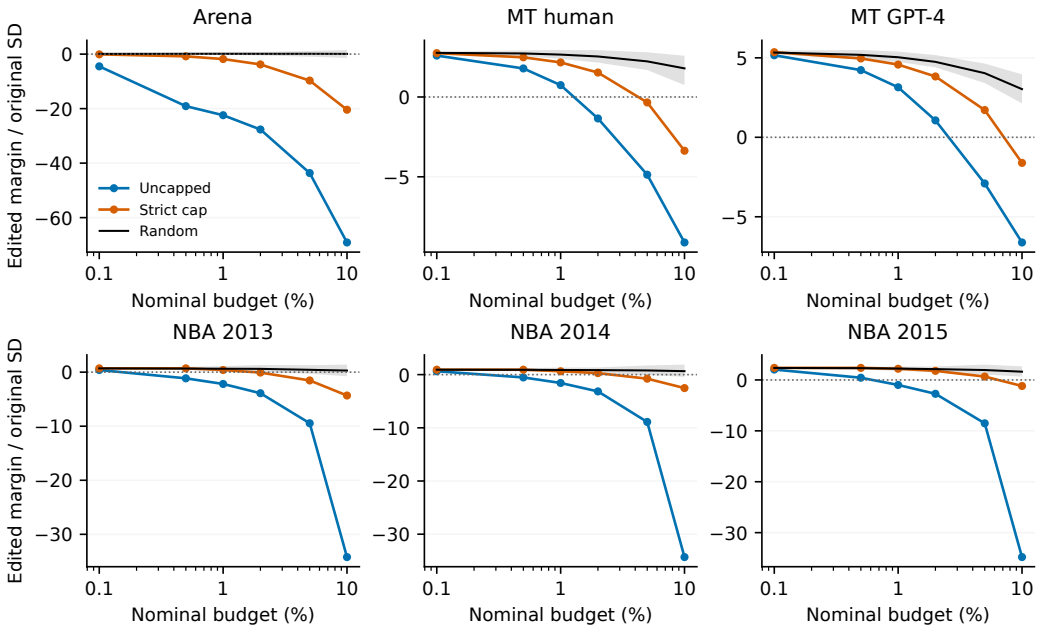


Figure 2: Budget curves for the fixed original top pair. Vertical scale is margin/original bootstrap SD, with different ranges across panels. Horizontal scale is logarithmic nominal global budget; exact integer counts and strict caps are recomputed at each point. Shading is the empirical 95% random-subset range. Crossing zero is a point-order reversal, not a significance reversal or a minimum-edit certificate.

The sports contrast has a different mechanism. Thirteen global edits correspond to roughly a tenth or more of one team's observed schedule, whereas strict 1% incident caps leave most teams with zero capacity. Equation (3) permits at most one, two, and one edits in the respective seasons; the greedy sets attain these count bounds. Smaller capped effects here do not establish that equally many dispersed edits would be harmless: equally many edits are not feasible under that restriction. The cap-plus-one check admits additional edits by *changing* the restriction, and is labeled accordingly (Appendix D).

The integer-linear check attains solver-reported optimality in all tested conditions and gives closely matching capped refit margins. It rules out a large greedy loss in the *tested first-order surrogate*; it does not establish a global optimum of the nonlinear perturbation problem. Figure 2 reports complete budget curves rather than selecting a favorable threshold.

5.2 Point flips, random noise, and original uncertainty differ

Arena's original margin is only 0.004, with percentile interval $[-0.103, 0.106]$. The original leader remains first in only 52% of original-data bootstrap replicates. Uniform 1% random reversals switch the pair in 35% of runs, while uncapped targeted reversals produce a much larger score change. A random or capped point flip therefore does not, by itself, destroy a well-established original separation. It can coexist with a valid warning that a sufficiently targeted edit changes scores substantially.

In the other full cohorts, random 1% reversals occur in zero or one of 100 trials; zero observed trials is not a zero-probability conclusion. Targeted reductions are substantially larger than random mean reductions (Figure 1), including where they fail to reverse the pair. The NBA2015 unperturbed interval excludes zero under the game-bootstrap approximation, yet its uncapped 1% witness reverses the pair. Conversely, MT's larger margins can survive an adversarial reduction of more than two original SDs. A flip count alone misses both distinctions.

Exploratory end-to-end resampling retains the large contrast between uncapped and capped reductions but changes the interpretation of individual reversals (Figure 4, Table 13). In particular, Arena's capped edited-margin percentile interval still crosses zero. The selected sets differ across resamples;

Table 3: Common MT cells, equal cell weighting, same fixed GPT-4-versus-Claude target. U and C margins follow nominal 1% *cell* reversals. Cell costs are not interchangeable with full-release judgment costs. Original intervals use synchronized question bootstrap draws.

Equal-weight panel	m_0	Original interval	m_U	m_C
MT human (cells)	0.293	[0.084, 0.540]	0.030	0.218
MT GPT-4 (cells)	0.772	[0.488, 1.099]	0.428	0.670

these distributions are neither repeated attempts at one fixed attack nor guarantees about an unknown population.

5.3 Matched judges: separation is not the same as response

The matched panels have exactly the same cells and endpoint degrees. GPT-4's original GPT-4-versus-Claude margin exceeds the human margin by 0.479, with paired question-bootstrap interval $[0.173, 0.795]$ (Table 3). Both retain positive margins under the 1% global edit procedure, although the human panel is much closer to a reversal. Their standardized uncapped reductions, 2.211 and 2.050, are similar in magnitude. This supports a distinction between initial separation and response to a particular perturbation, not a claim that a model judge is more accurate, less biased, or generally safer.

5.4 Specification checks expose non-universality

The main full-cohort reversal pattern persists under the two ridge alternatives, but this is not invariance to the entire modeling pipeline. Dropping exact ties already switches the original Arena top pair *before* editing; attributing that switch to the edit would be incorrect. Excluding inconsistent GPT-4 outcomes and restricting human judge sources change both initial margins and the effective denominators (Tables 9–10). Whole-question reversals produce another response curve because a question spans many correlated judgments, rather than being an interchangeable vote (Table 11).

Static and batch-adaptive outcomes, cap rounding, and integer-linear selection are reported together in Table 8. Exact-refit checks are essential: the Arena 1% deletion control predicts a reduction of about 0.764 but actually reduces the margin by about 2.281. This is finite-perturbation nonlinearity, despite accurate infinitesimal derivatives. Appendix E gives the complete linear-versus-refit diagnostic and grid witnesses without claiming optimal thresholds.

6 Discussion and limitations

Scope of the finding. The supported claim is not that leaderboards are universally fragile or robust. It is that a global fraction does not specify exposure, and small point-rank changes have different interpretations depending on initial separation and feasible edits. These statements follow from actual counterfactual refits and explicit denominators. The strict-cap comparison is an audit of a different admissible set, not evidence that platforms enforce such caps or that they remove all dangerous perturbations.

Statistical limits. The top pair is selected from the same data and held fixed; intervals are descriptive, not winner-selection-adjusted inference. Prompt-string clustering does not recover user dependence, question clustering does not model shared judges across questions, and NBA game resampling does not capture serial, team, home-court, or season dynamics. Static BT may be misspecified, especially for changing sports strengths or nontransitive model preferences. A ridge-finite fit need not be a well-specified generative model. No cross-domain ordering should be interpreted causally, and six full cohorts do not estimate a population prevalence of vulnerable leaderboards.

Perturbation and computational limits. We assume offline access to the observed schedule and outcomes. Real manipulation has costs, detection, and access restrictions absent here. Tie reversals, deletions, question edits, and averaged-cell reversals have different semantics. Same-λ sum-loss fits have different relative regularization across sample sizes. Static influence and its integer surrogate do

not solve the nonlinear minimum-edit problem. Small-budget zeros can be arithmetic restrictions rather than statistical robustness. Optional matched-exposure Arena subsampling was not run; no equal-exposure causal contrast is claimed. The bounded literature protocol prevents an exhaustive claim of originality.

6.1 Broader impact and research process

Exposure-aware reporting can help readers avoid unsupported selection and marketing claims. Conversely, edit-selection code could be repurposed to manipulate evaluations or to exaggerate the significance of a harmless near-tie. We release an offline audit with exposure and uncertainty diagnostics, no live-voting integration, no account identification, and no raw conversations in the code archive. These choices reduce operational and privacy risks but are not complete misuse safeguards. No new participants were recruited or identified; all analyses use public licensed historical records. No new institutional review was obtained, and no unverified approval or exemption is asserted. Substantial LLM assistance and the restricted-access protocol are disclosed in the appendix.

7 Conclusion

A useful stability report should state *what* is edited, *whose* evidence is affected, *how many* edits are actually feasible, and *which* conclusion changes relative to original uncertainty. On the tested snapshots, global influence-selected reversals, strict exposure caps, and random errors lead to materially different interpretations. Reporting only a small global fraction or a flipped first-place label omits those distinctions. Our evidence is a set of reproducible conditional witnesses and controls, not a universal robustness verdict.

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A Objective, derivatives, and counting identities

Write $L(\theta; y, w) = \sum_r w_r[\log(1 + \exp(x_r^\top \theta)) - y_r x_r^\top \theta] + \lambda\|\theta\|^2/2$, with all original $w_r = 1$ and $\lambda > 0$. The gradient and Hessian are

$$g(\theta) = \sum_r w_r x_r (p_r - y_r) + \lambda\theta, \tag{5}$$

$$H(\theta) = \sum_r w_r p_r (1 - p_r) x_r x_r^\top + \lambda I. \tag{6}$$

For nonnegative weights, H is positive definite. The loss is smooth and coercive because of the positive quadratic penalty, so a unique finite minimizer exists. Summing the gradient coordinates gives $\lambda \sum_i \widehat{\theta}_i = 0$. This justifies centered coordinates without an arbitrary reference entrant.

For a single record, continuously interpolate its outcome as $y_r(t) = y_r + t(1 - 2y_r)$, holding weights fixed. Differentiating $g(\widehat{\theta}(t), t) = 0$ at $t = 0$ gives

$$H\frac{d\widehat{\theta}}{dt} + x_r(2y_r - 1) = 0, \qquad -\frac{dm}{dt} = c^\top H^{-1} x_r (2y_r - 1). \tag{7}$$

For deletion, use $w_r(t) = 1 - t$ and fixed response. The corresponding derivative is

$$H\frac{d\widehat{\theta}}{dt} - x_r(p_r - y_r) = 0, \qquad -\frac{dm}{dt} = c^\top H^{-1} x_r (y_r - p_r). \tag{8}$$

The actual discrete operation uses $t = 1$, for which this derivative need not approximate the full refit accurately. Positive-score sorting optimizes only the additive first-order objective. The ridge term remains in the Hessian and is not removed or rescaled after deletion. These standard implicit derivatives are implementation identities, not a novel influence theorem.

For exposure, $\sum_i n_i = 2N$ and $\sum_i k_i(S) = 2|S|$, because each retained comparison has exactly two distinct endpoints. Substitution proves Eq. (2); a weighted average cannot exceed the maximum of its terms. Summing strict capacities and dividing by two proves Eq. (3); the integer floor follows because $|S|$ is integral. This count bound does not depend on any fitted score or attack heuristic. It is generally not tight when endpoint pairs, beneficial outcomes, or other constraints limit feasibility.

Numerical implementation. We aggregate counts and response sums for each canonical unordered pair; this is exact for Eq. (1), including fractional responses. Newton steps solve the dense positive-definite system using SciPy, with backtracking and a default maximum gradient target 10^{-8}. A fit fails if its final gradient exceeds 2×10^{-6} or has a nonfinite state; no such production failure occurred. Canonical orientation is checked at the data boundary because lower-triangle inputs would otherwise be silently omitted by upper-triangle sufficient statistics. Analytical two-player and finite-difference gradient, Hessian, outcome, and deletion tests are supplied, along with integer-surrogate brute force on a toy example. The toy fixture is a unit test, not experimental synthetic evidence.

For the integer-linear control, one bounded integer variable counts selected records in an observed (a, b, y) class; all members have equal influence. Endpoint incidence constraints and a global cardinality constraint are linear. SciPy's mixed-integer solver uses a 60-second limit and relative gap target 10^{-9}; every recorded condition terminates with status zero and reported gap zero. Refitted margins remain feasible nonlinear witnesses only. Different equal-objective class allocations can produce different discrete sets or counts.

B Data snapshots, provenance, and edit semantics

The data owners are LMArena/LMSYS for Arena, the MT-Bench authors for the judgment release, and FiveThirtyEight for the NBA archive. We downloaded only the fixed public snapshots specified in the task. Table 4 records the commits and terms; the code's `artifacts/data_manifest.json` records exact resolve URLs, bytes, and SHA256 values for all four source files. Three supplied hashes are checked before parsing; the fourth (GPT-4 parquet) is computed and recorded. Raw data remain in `data/` and are excluded from the reproduction archive.

Table 4: Source snapshots. Dataset cards and repository license were fetched directly and retained in the provenance bundle. No third-party rehosting or restricted dataset is needed.

Source	Public repository and fixed commit	License
Arena55k	https://huggingface.co/datasets/lmarena-ai/arena-human-preference-55k; 18c298340948c0e7f7727399fd459cca6ce0ca6f	Apache-2.0
MT human / GPT-4	https://huggingface.co/datasets/lmsys/mt_bench_human_judgments; f7d2896d2cc5d80f8b55c2bbc722613555233c25	CC-BY-4.0
NBA historical	https://github.com/fivethirtyeight/data; 4c1ff5e3aef1816ae04af63218015066e186c147; nba-elo/nbaallelo.csv	CC-BY-4.0

Arena uses all eligible rows and all observed models, with no outcome-based top-model trimming. Winner columns are checked to sum to one; self comparisons and invalid winners would be excluded and counted. The retained full snapshot has no such exclusions or duplicate record IDs. Prompt content is used only for exact-string hashing to define resampling clusters; no conversations are quoted or redistributed. The term "55k" is the public dataset label rather than the exact retained count in Table 1.

MT processing maps observed winner labels to $1, 0, 0.5$ and canonicalizes model orientation before aggregation. There are 380 GPT-4 records labeled `tie (inconsistent)`, distinct from 220 explicit ties; the main fit includes both as half outcomes. The human source restrictions select the observed `expert_` versus `author_` label prefix without inferring identities. `mt_human_cell_provenance.csv` and its GPT-4 counterpart retain question, turn, unordered pair, multiplicity, number of distinct judges, and source row indices, sufficient to reconstruct each aggregate without redistributing conversations. Each common cell is weighted equally, regardless of its human judgment multiplicity. Matched-cell response reversals invert the entire mean; their nominal cell budgets are not raw-vote budgets.

For NBA, filter league NBA and the three fixed `year_id` values, then retain `_iscopy==0`. Game IDs are verified unique. A game outcome is one binary comparison; the mirrored opponent row is not a second observation. Historical records are used as a cross-domain test of a static score aggregation, not to infer current team ability, home advantage, or misconduct.

C Uncertainty details and additional controls

Each original-data replicate samples the same number of clusters as observed, with replacement, and assigns every row its cluster multiplicity as a likelihood weight. The pair remains fixed to the full-data leader/runner. The base seed is 20261104; original and exploratory bootstraps use seed base $+100000 + b$, and random controls use base $+r$ or base $+200000 + r$. Random sets at different nominal budgets share a permutation within a seed and are therefore nested, not independent across the budget curve. This coupling is deliberate and is not used for multiplicity-unadjusted significance tests.

Table 5: Uncertainty at nominal 1% on full cohorts. s_0: original margin SD. Win retention: original leader remains first under cluster bootstrap. Flip: random-subset pair-reversal frequency with Wilson 95% interval (100 draws). Last column: empirical 2.5–97.5% random edited-margin range, not an interval for an estimated mean.

Cohort	s_0	Win retain%	Flip%	Wilson interval%	Random margin range
Arena	0.055	52.0	35	[26.4, 44.7]	[-0.018, 0.038]
MT human	0.136	100.0	0	[0.0, 3.7]	[0.325, 0.401]
MT GPT-4	0.153	100.0	0	[0.0, 3.7]	[0.726, 0.826]
NBA 2013	0.324	64.7	1	[0.2, 5.4]	[0.069, 0.364]
NBA 2014	0.318	78.3	0	[0.0, 3.7]	[0.152, 0.426]
NBA 2015	0.314	96.7	0	[0.0, 3.7]	[0.568, 0.825]

The count-matched control samples exactly as many non-tie records as the corresponding capped greedy set, without enforcing its endpoint restriction. Table 6 reports both supplementary random controls. The cap-feasible control shuffles priorities and accepts records greedily if both endpoint slots remain. It is a heuristic feasible sampler, not a uniform distribution over feasible sets or an equal-count control. Reporting their actual mean counts and exposure avoids conflating those roles.

Table 6: Supplementary random controls at nominal 1%. D uses the same original SD as the primary analysis. Cap-feasible draws can underfill; count-matched draws need not satisfy the caps.

Cohort	Random control	Mean edits	Mean 100ρ	Mean D	95% subset range
Arena	Count-matched	555.0	2.07	-0.014	[-0.430, 0.413]
Arena	Cap-feasible	557.0	1.00	-0.020	[-0.464, 0.365]
MT human	Count-matched	29.0	1.19	0.085	[-0.145, 0.320]
MT human	Cap-feasible	31.2	0.99	0.107	[-0.129, 0.338]
MT GPT-4	Count-matched	24.0	1.41	0.276	[-0.029, 0.594]
MT GPT-4	Cap-feasible	23.4	1.00	0.247	[0.003, 0.417]
NBA 2013	Count-matched	1.0	1.19	0.002	[-0.138, 0.161]
NBA 2013	Cap-feasible	1.0	0.99	0.037	[-0.311, 0.309]
NBA 2014	Count-matched	2.0	1.31	0.008	[-0.140, 0.158]
NBA 2014	Cap-feasible	2.0	0.99	-0.008	[-0.300, 0.298]
NBA 2015	Count-matched	1.0	1.19	0.007	[-0.144, 0.188]
NBA 2015	Cap-feasible	1.0	0.98	0.072	[-0.182, 0.175]

Matched contrast. The common panels contain the same question IDs and cell order; matching is by question, turn, and unordered pair. Synchronized bootstrap question draws therefore support a paired difference of the *same* GPT-4-versus-Claude margin. This difference does not establish evaluator accuracy: there is no external ground truth for which model should have the larger gap. Unmatched full releases are not substituted for this paired target.

D Cap feasibility and specification stress tests

Table 7 separates endpoint-slot arithmetic from the selected procedure's behavior. Table 8 compares the tested selection and rounding variants.

Table 7: Primary strict-cap arithmetic. Eligible rows are non-ties whose two endpoints have positive capacities, before considering their influence signs. Count upper bound is Eq. (3); edges may reduce feasibility further. In NBA this upper bound already explains the one/two/one selected counts.

Cohort	Global budget	Zero-cap entrants	Eligible rows	Count upper bound	Greedy edits
Arena	574	0	39716	559	555
MT human	33	0	2575	32	29
MT GPT-4	24	0	1800	24	24
NBA 2013	13	27	21	1	1
NBA 2014	13	26	33	2	2
NBA 2015	13	28	8	1	1

Table 8: Exact edited fixed-pair margins at nominal 1%. Adaptive U uses at most ten batches; Integer C optimizes the additive influence surrogate under strict caps. Cap+1 replaces every endpoint limit by $\lfloor \alpha n_i \rfloor + 1$ while keeping the original global budget. Its larger realized ρ demonstrates a changed restriction, not equivalent rounding.

Cohort	Static U	Adaptive U	Greedy C	Integer C	Cap+1	$100\rho_{+1}$
Arena	-1.244	-1.244	-0.100	-0.101	-0.106	2.00
MT human	0.101	0.101	0.294	0.294	0.286	1.08
MT GPT-4	0.482	0.482	0.700	0.700	0.686	1.12
NBA 2013	-0.707	-0.789	0.128	0.128	-0.022	2.00
NBA 2014	-0.500	-0.576	0.200	0.200	0.088	1.98
NBA 2015	-0.309	-0.342	0.689	0.689	0.560	1.96

Table 9: Uncapped 1% checks retain the original full-cohort pair. Each specification has its own N, unedited fit, degrees, and floor budget. A "Base reversed" flag means the unedited altered specification already reverses that pair. NBA tie dropping changes no rows; it is intentionally a null ablation.

Cohort	Specification	N	Unedited margin	Edited margin	Base reversed
Arena	Ridge 0.01	57477	0.004	-1.245	0
Arena	Ridge 1	57477	0.008	-1.231	0
Arena	Drop exact ties	39716	-0.033	-6.184	1
MT human	Ridge 0.01	3355	0.378	0.101	0
MT human	Ridge 1	3355	0.376	0.101	0
MT human	Drop exact ties	2575	0.532	0.208	0
MT GPT-4	Ridge 0.01	2400	0.821	0.482	0
MT GPT-4	Ridge 1	2400	0.809	0.476	0
MT GPT-4	Drop exact ties	1800	1.100	0.563	0
NBA 2013	Ridge 0.01	1314	0.229	-0.713	0
NBA 2013	Ridge 1	1314	0.219	-0.652	0
NBA 2013	Drop exact ties	1314	0.228	-0.707	0
NBA 2014	Ridge 0.01	1319	0.297	-0.504	0
NBA 2014	Ridge 1	1319	0.279	-0.467	0
NBA 2014	Drop exact ties	1319	0.295	-0.500	0
NBA 2015	Ridge 0.01	1311	0.750	-0.312	0
NBA 2015	Ridge 1	1311	0.693	-0.285	0
NBA 2015	Drop exact ties	1311	0.744	-0.309	0

Whole-question edits reverse every non-tie judgment in a selected MT question, across all observed pairs and turns. Question influence is the sum of row scores, and questions with beneficial total influence are selected in descending order. The question budget is separate from the changed-judgment budget; ties are touched but unchanged. These group edits are not constrained by the

Table 10: Observed judge-source and inconsistency-label sensitivities. U/C denote uncapped/strict capped outcome reversals. Different restrictions change the target record population, not merely an attack algorithm. They are exploratory descriptive checks.

Cohort	Restriction	N	m_0	U1%	C1%	U5%	C5%
MT human	experts only	2668	0.403	0.144	0.319	-0.640	-0.026
MT human	authors only	687	0.255	-0.069	0.198	-0.838	-0.163
MT GPT-4	exclude inconsistent	2020	0.992	0.548	0.833	-0.774	0.264

row-level endpoint caps. All costs and final margins are reported rather than equating a question with one vote.

Table 11: MT whole-question reversals. Fractions use all 80 questions or the original full-release row count, respectively. "Changed judgments" excludes exact ties; touched-row counts are also retained in the source CSV.

Cohort	Questions	Question%	Changed judgments	Row%	Margin
MT human	1	1.25	39	1.16	0.274
MT human	2	2.50	88	2.62	0.173
MT human	4	5.00	173	5.16	0.036
MT human	8	10.00	320	9.54	-0.209
MT GPT-4	1	1.25	12	0.50	0.714
MT GPT-4	2	2.50	26	1.08	0.615
MT GPT-4	4	5.00	60	2.50	0.439
MT GPT-4	8	10.00	120	5.00	0.147

E Approximation accuracy and observed grid witnesses

Figure 3 audits finite-budget approximation errors; Table 12 lists the first observed grid witnesses without claiming minimum budgets.

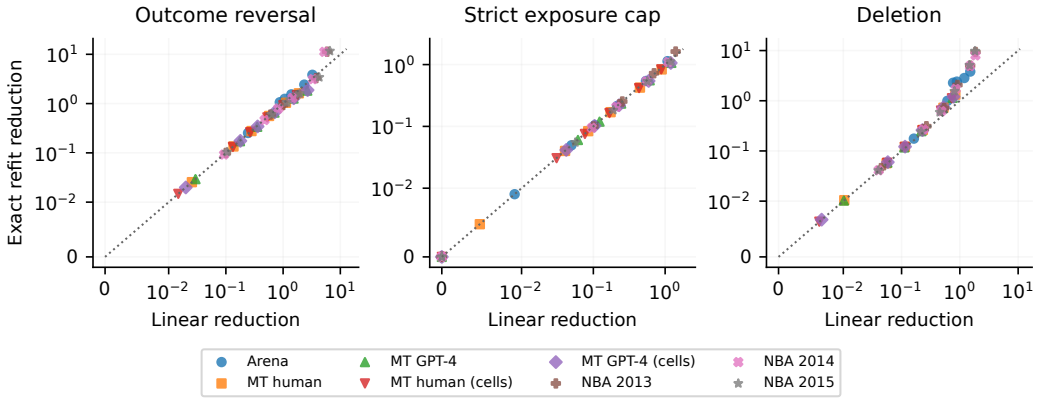


Figure 3: Exact-refit versus first-order predicted reductions for every nonzero grid budget and all eight cohorts. Both axes are symlog, linear within 0.01. The dotted line is equality, not a regression. Large finite-budget discrepancies, especially for deletions, preclude using a linearized ranking change as a certificate.

Table 12: First *observed* pair reversal on the fixed nominal grid, not the minimum necessary budget. No intermediate budgets or alternative nonlinear edit sets are searched. Matched rows are aggregate-cell units, unlike full-release judgment/game units.

Cohort	Uncapped flip	Capped flip	Deletion
Arena	0.1%	0.1%	0.1%
MT human	2%	5%	5%
MT GPT-4	5%	10%	10%
MT human (cells)	2%	5%	5%
MT GPT-4 (cells)	5%	10%	10%
NBA 2013	0.5%	2%	0.5%
NBA 2014	0.5%	5%	1%
NBA 2015	1%	10%	2%

All deterministic primary and supplementary edit sets have compressed NPZ witnesses containing Unicode row IDs, canonical endpoints, original responses, and selected indices. They are loaded with `allow_pickle=False`. The validator reconstructs the appropriate tie/source-restricted data specification and refits from a cold start. It also checks unique indices, exact original responses, nominal global counts, and strict endpoint limits. The current audit replays 338 deterministic witnesses; the maximum recorded margin discrepancy is below 10^{-7}. Numerical fit counts and gradients are separately reconciled for the primary, supplemental, and exploratory-bootstrap runs in `validation.json`. Random sets and bootstrap weights are regenerated from saved seeds and immutable data ordering rather than stored as fabricated witnesses.

F Exploratory end-to-end bootstrap

This analysis was added after inspecting the production endpoints, and is explicitly separated from the locked primary plan. For each of 300 cluster draws, expand rows according to cluster multiplicity, refit the unedited data, recompute N, entrant degrees, the floor budget, influence ordering, and strict capacities, then select and refit new sets at 1%. The original full-data contrast is retained, even if already reversed in the unedited resample. All fits use the same ridge as the primary analysis. The intervals describe the entire data-resampling/selection/refitting procedure, not one fixed selected set. No extra replicates were added to shift a percentile endpoint.

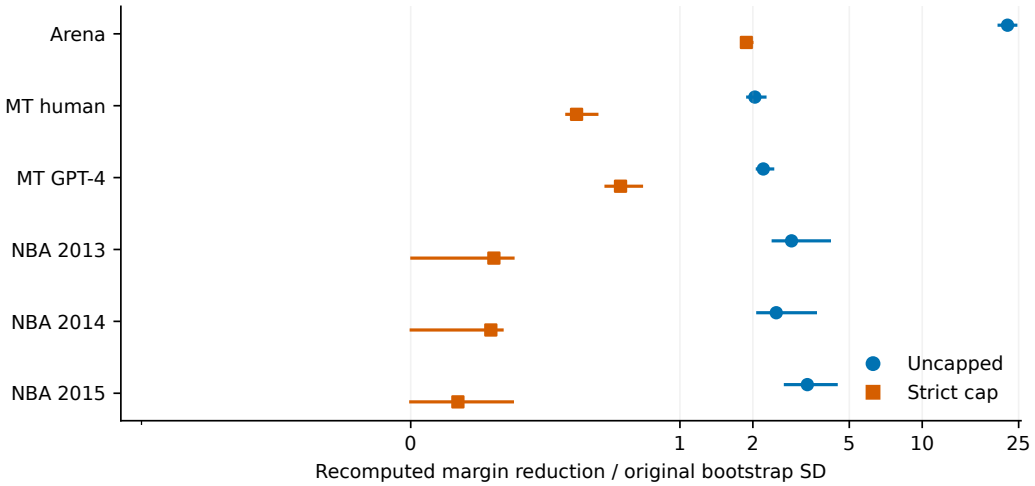


Figure 4: Exploratory end-to-end effect distributions. Points are original-data reductions; whiskers are percentile reduction ranges from recomputed procedures, divided by the original fixed SD. Symlog horizontal scale is linear within one unit. These are not selection-adjusted intervals or post-edit significance tests.

Table 13: Raw-margin percentile intervals for the exploratory recomputed procedure. Reversed% is the fraction of edited replicate margins below zero, not a frequentist guarantee or a conditional causal effect of editing. It can include resamples with an already reversed original pair.

Cohort	Selector	Reduction interval	Edited-margin interval	Reversed%
Arena	U	[1.152, 1.350]	[-1.353, -1.140]	100.0
Arena	C	[0.101, 0.110]	[-0.209, 0.001]	97.0
MT human	U	[0.259, 0.305]	[-0.176, 0.382]	22.7
MT human	C	[0.079, 0.094]	[0.021, 0.574]	1.0
MT GPT-4	U	[0.319, 0.370]	[0.215, 0.754]	0.0
MT GPT-4	C	[0.111, 0.131]	[0.430, 0.991]	0.0
NBA 2013	U	[0.785, 1.339]	[-1.528, -0.190]	99.3
NBA 2013	C	[0.001, 0.123]	[-0.486, 0.772]	33.7
NBA 2014	U	[0.667, 1.154]	[-1.396, 0.106]	95.0
NBA 2014	C	[0.001, 0.108]	[-0.427, 0.853]	19.3
NBA 2015	U	[0.855, 1.385]	[-1.070, 0.215]	87.3
NBA 2015	C	[0.000, 0.119]	[0.109, 1.326]	1.0

G Reproducibility, compute, and failures

The code archive contains `src/`, `tests/`, the pinned `requirements.txt`, `reproduce.py`, and a README with the data commands and script-to-table/figure map. Run `python reproduce.py` from the workspace root after installing the requirements; cached original data are checksum-verified. Individual commands are:

```
python src/download_data.py
python -m pytest tests -q
python src/run_experiments.py --bootstrap 300 --random 100
python src/run_stress.py --replicates 100
python src/bootstrap_sensitivity.py
python src/validate_results.py
python src/summarize.py
python src/plot_figures.py
python src/complete_checklist.py
latexmk -pdf -cd paper/main.tex
```

Set `OPENBLAS_NUM_THREADS=1`, `OMP_NUM_THREADS=1`, and `MKL_NUM_THREADS=1` before execution. The numerical environment is Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6, Matplotlib 3.11.2, PyArrow 26.0.1, pytest 9.2.0, and pypdf 6.21.4. PDF compilation used pdfTeX 1.40.26 (TeX Live 2025/dev/Debian) and the unmodified official NeurIPS 2026 style. The runtime exposes 40 logical CPUs but each numerical process requests one thread; the CPU model and host memory are not exposed through the permitted inspection interface. This does not imply the experiments used 40 cores.

The final replay measured 17.3 seconds for the core experiment process, 17.2 seconds for supplemental controls, and 28.9 seconds for the exploratory bootstrap. The complete download-verification, test, analysis, validation, and rendering sequence took 80.4 seconds before PDF compilation. Peak child resident memory across these stages was 349.5 MiB (Linux cumulative child-process maximum RSS). These are measured times on the provided shared host, not guarantees for another machine. Original pilot and dependency-recovery costs are additional and are logged separately.

Every scientific CSV can be regenerated without a GPU or paid API. `summarize.py` produces all numerical tables and the claim ledger; `plot_figures.py` reads those CSVs to produce every figure. `results_macros.tex` is generated, not a source of invented result numbers. Sufficient-statistic fitting avoids storing a dense $N \times P$ design matrix. Download and TeX/dependency installation time are additional and network-dependent; the budget is comfortably below one CPU week for these snapshots, not a complexity guarantee for larger systems.

Failures and abandoned branches. The first NPZ replay failed because object-typed string IDs required unsafe pickle loading; IDs were changed to Unicode and all witnesses regenerated. No unsafe deserialization workaround was used. Initial PDF-extraction dependency recovery and a legacy template URL failure are logged separately; the official 2026 bundle was then obtained from the conference CFP. A blocked system-path CPU-model probe was not bypassed. Optional Arena matched-exposure subsampling was deferred. We did not train manipulation classifiers, call live judges, reproduce a published minimum-edit number, or run a causal evaluator-design experiment. Unsuccessful or unperformed branches are not represented as positive evidence.

H LLM assistance, licensing, and access boundaries

An LLM-based research assistant, operating as EvoScientist, autonomously selected the study design, consulted the allowed literature, wrote and tested analysis code, interpreted measured results, and drafted the manuscript. Subagents supplied literature notes, stage critique, and manuscript self-review. The runtime did not expose a verified underlying model identifier; none is invented here. LLM reasoning is not an empirical measurement or mathematical certification: all reported data values, tables, and plotted points are computed by the included CPU programs. No paid inference API, newly generated judge labels, or GPU was used. A human scientific and legal review remains appropriate before actual submission.

New analysis code is supplied under the MIT license; third-party datasets, official style files, and consulted papers retain their original terms. Derived audit summaries and row-ID witnesses do not relicense underlying assets. Raw conversations, all original data files, and third-party paper PDFs are excluded from distribution archives; the source/provenance manifests identify how to obtain them legitimately. The official 2026 checklist is retained with all sixteen answers and justifications.

H.1 Restricted-access protocol

The task specified an unidentified prior paper and prohibited access to it and to pages discussing it, while explicitly permitting a list of starter references. No attempt was made to infer its identity. Literature access was limited to exact permitted arXiv IDs and their own PDFs; no keyword search, citation traversal, related-paper service, review page, or repository search was used. Targeted sections actually read, downloaded versions, hashes, and access failures appear in `artifacts/literature_notes.md` and `access_and_failure_log.md`. This bounded protocol prevents exhaustive novelty clearance. No recognized prohibited source was encountered outside the explicitly permitted set. References printed within permitted PDFs were not followed or used as independently read sources. Later data-card/license, official ethics, and template accesses were exact allowed asset/compliance pages and are logged; no linked further-reading papers were fetched.

NeurIPS Paper Checklist

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   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 6 discusses target selection, dependence, model misspecification, edit semantics, integer caps, and restricted novelty clearance.

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   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
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3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Appendix A gives the complete objective, positive-ridge assumption, implicit derivative calculation, and elementary exposure-count identities. No novel robustness theorem or certified nonlinear bound is claimed.

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- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
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Justification: Section 4 and Appendix G specify all preprocessing, seeds, optimizers, snapshots, and commands; the accompanying code recreates every figure and table.

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  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The accompanying reproducibility archive contains all analysis code and exact public-data download commands, with versions and SHA256 hashes (Appendix G); raw conversations are not redistributed.

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   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
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   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Sections 3–4 specify the fixed-snapshot estimand, ridge, tie policy, budgets, bootstrap clustering, and random controls. There is no predictive train/test split or learned attack classifier.

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Tables 1 and 5, Figures 2 and 4, and Section 3.4 separate cluster-bootstrap intervals, random-subset variability, and Wilson Monte Carlo intervals; selection/dependence caveats are explicit.

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- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
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